

# Лабораторная работа №4

## Презентация

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1 Информация

2 Выполнение лабораторной работы

# Раздел 1

## Информация

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## Раздел 2

### Выполнение лабораторной работы

# Запускаем наш сервер

```
Попробуйте новую кроссплатформенную оболочку PowerShell (https://aka.ms/pscore6)

PS C:\Users\Ванек> cd C:\work\iazubov\vagrant
PS C:\work\iazubov\vagrant> vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
    server: Adapter 1: nat
    server: Adapter 2: intnet
==> server: Forwarding ports...
    server: 22 (guest) => 2222 (host) (adapter 1)
==> server: Running 'pre-boot' VM customizations...
==> server: Booting VM...
==> server: Waiting for machine to boot. This may take a few minutes...
    server: SSH address: 127.0.0.1:2222
    server: SSH username: vagrant
    server: SSH auth method: password
==> server: Machine booted and ready!
==> server: Checking for guest additions in VM...
    server: No guest additions were detected on the base box for this VM! Guest
    server: additions are required for forwarded ports, shared folders, host only
    server: networking, and more. If SSH fails on this machine, please install
    server: the guest additions and repackage the box to continue.
    server:
    server: This is not an error message; everything may continue to work properly,
    server: in which case you may ignore this message.
==> server: Setting hostname...
==> server: Configuring and enabling network interfaces...
==> server: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> server: flag to force provisioning. Provisioners marked to run always will still run.
==> server: Running provisioner: common hostname (shell)...
```

# Установка веб сервера

```
root@server:~ - sudo -i

vagrant@server:~$ sudo -i
root@server:~# LANG=C yum grouplist
Rocky Linux 10 - BaseOS                12 kB/s | 4.3 kB    00:00
Rocky Linux 10 - BaseOS                17 MB/s | 31 MB    00:01
Rocky Linux 10 - AppStream              15 kB/s | 4.4 kB    00:00
Rocky Linux 10 - AppStream              2.3 MB/s | 2.7 MB   00:01
Rocky Linux 10 - Extras                 9.8 kB/s | 3.1 kB    00:00
Available Environment Groups:
  Server
  Minimal Install
  Workstation
  Custom Operating System
  Virtualization Host
Installed Environment Groups:
  Server with GUI
Installed Groups:
  Container Management
  Headless Management
Available Groups:
  Legacy UNIX Compatibility
  Smart Card Support
  Console Internet Tools
  Development Tools
  .NET Development
  Graphical Administration Tools
  Network Servers
  RPM Development Tools
  Scientific Support
  Security Tools
  System Tools
root@server:~# dnf -y groupinstall "Basic Web Server"
```

# Основной файл httpd.conf

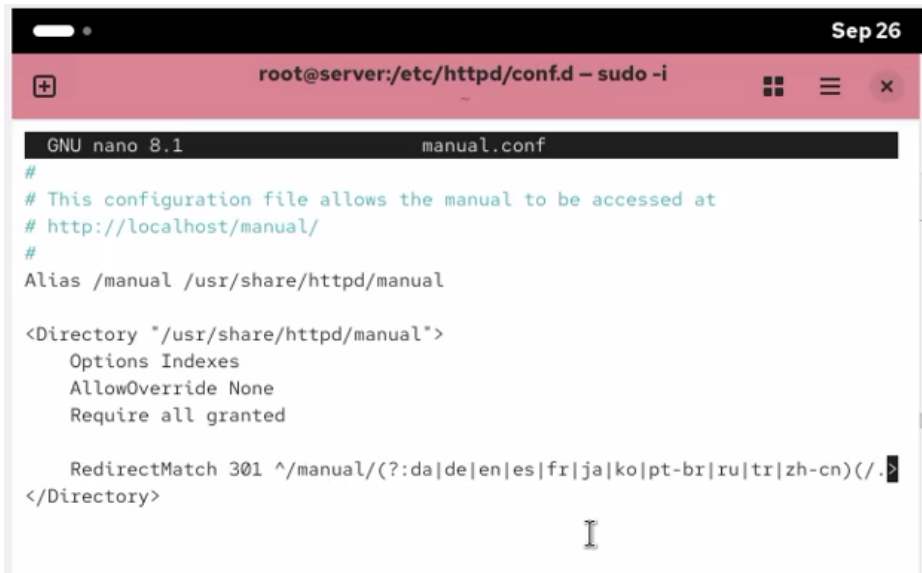
```
root@server:/etc/httpd/conf -- sudo -i
GNU nano 8.1 httpd.conf
# directive tells the module where the hint definitions are located.
#
MIMEMagicFile conf/magic
</IfModule>

#
# Customizable error responses come in three flavors:
# 1) plain text 2) local redirects 3) external redirects
#
# Some examples:
#ErrorDocument 500 "The server made a boo boo."
#ErrorDocument 404 /missing.html
#ErrorDocument 404 "/cgi-bin/missing_handler.pl"
#ErrorDocument 402 http://www.example.com/subscription_info.html
#

#
# EnableMMAP and EnableSendfile: On systems that support it,
# memory-mapping or the sendfile syscall may be used to deliver
# files. This usually improves server performance, but must
# be turned off when serving from networked-mounted
# filesystems or if support for these functions is otherwise
# broken on your system.
# Defaults if commented: EnableMMAP On, EnableSendfile Off
#
#EnableMMAP off
EnableSendfile on

# Supplemental configuration
#
# Load config files in the "/etc/httpd/conf.d" directory, if any.
IncludeOptional conf.d/*.conf
```





```
root@server:/etc/httpd/conf.d - sudo -i
GNU nano 8.1 manual.conf
#
# This configuration file allows the manual to be accessed at
# http://localhost/manual/
#
Alias /manual /usr/share/httpd/manual

<Directory "/usr/share/httpd/manual">
    Options Indexes
    AllowOverride None
    Require all granted

    RedirectMatch 301 ^/manual/(?:da|de|en|es|fr|ja|ko|pt-br|ru|tr|zh-cn)(/.)>
</Directory>
```



The screenshot shows a terminal window with a title bar indicating the date as 'Sep 26'. The terminal prompt is 'root@server:/etc/httpd/conf.d - sudo -i'. The nano editor is open, editing the file 'ssl.conf'. The editor's status bar at the top shows 'GNU nano 8.1' and the filename 'ssl.conf'. The content of the file is as follows:

```
#
# When we also provide SSL we have to listen to the
# standard HTTPS port in addition.
#
Listen 443 https

###
###  SSL Global Context
###
###  All SSL configuration in this context applies both to
###  the main server and all SSL-enabled virtual hosts.
###

#  Pass Phrase Dialog:
#  Configure the pass phrase gathering process.
#  The filtering dialog program ('builtin' is a internal
#  terminal dialog) has to provide the pass phrase on stdout.
SSLPassPhraseDialog exec:/usr/libexec/httpd-ssl-pass-dialog

#  Inter-Process Session Cache:
#  Configure the SSL Session Cache: First the mechanism
#  to use and second the expiring timeout (in seconds).
SSLSessionCache          shmcb:/run/httpd/sslcache(512000)
SSLSessionCacheTimeout   300

#
# Use "SSLCryptoDevice" to enable any supported hardware
# accelerators. Use "openssl engine -v" to list supported
# engine names. NOTE: If you enable an accelerator and the
# server does not start, consult the error logs and ensure
# your accelerator is functioning properly.
#
```



The screenshot shows a terminal window with a black title bar containing a battery icon and the date "Sep 26". Below the title bar is a pink header bar with the text "root@server:/etc/httpd/conf.d - sudo -i" and window control icons. The main area of the terminal is a black box with the text "GNU nano 8.1 userdir.conf". The content of the file is displayed in a light blue monospace font. The file contains several comments explaining the UserDir module, followed by an `<IfModule mod_userdir.c>` block. Inside this block, there is a comment about disabling UserDir by default, followed by the line `UserDir disabled`. Below this, there is another comment about enabling requests to `/~user/` and a line `#UserDir public_html`. The block ends with `</IfModule>`. At the bottom of the file, there is a comment about controlling access to UserDir directories and an example `<Directory "/home/*/public_html">` block with the lines `AllowOverride FileInfo AuthConfig Limit Indexes`. A cursor is visible on the line containing `#UserDir public_html`.

```
#
# UserDir: The name of the directory that is appended onto a user's home
# directory if a ~user request is received.
#
# The path to the end user account 'public_html' directory must be
# accessible to the webserver userid. This usually means that ~userid
# must have permissions of 711, ~userid/public_html must have permissions
# of 755, and documents contained therein must be world-readable.
# Otherwise, the client will only receive a "403 Forbidden" message.
#
<IfModule mod_userdir.c>
#
# UserDir is disabled by default since it can confirm the presence
# of a username on the system (depending on home directory
# permissions).
#
UserDir disabled

#
# To enable requests to /~user/ to serve the user's public_html
# directory, remove the "UserDir disabled" line above, and uncomment
# the following line instead:
#
#UserDir public_html
</IfModule>

#
# Control access to UserDir directories. The following is an example
# for a site where these directories are restricted to read-only.
#
<Directory "/home/*/public_html">
    AllowOverride FileInfo AuthConfig Limit Indexes
```

# Настройка межсетевого экрана

```
n bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-ex
porter ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit
collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dh
cpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker-registry dock
er-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finge
r foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replicat
ion freeipa-trust ftp galera ganglia-client ganglia-master git gitea gpsd gra
fana gre high-availability http http3 https ident imap imaps iperf2 iperf3 ip
fs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect
kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-cont
rol-plane kube-control-plane-secure kube-controller-manager kube-controller-m
anager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kub
e-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-t
ls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matri
x mdns memcache minecraft minidlna mndp mongodb mosh mountd mpd mqtt mqtt-tls
ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns n
etdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-
imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmweb
apis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-
dhcp proxy-http ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius
radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-m
aster samba samba-client samba-dc sane settlers-history-collection sip sips s
limevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap sock
s spideroak-lansync spotify-sync squid ssdp ssh statsrv steam-lan-transfer st
eam-streaming stellaris stronghold-crusader stun stuns submission supertuxkar
t svdrp svn syncthing syncthing-gui syncthing-relay synergy syscomlan syslog
syslog-ng syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks tran
smission-client turn turns upnp-client vdsd vnc-server vrrp warpinator wbem-h
ttp wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host w
s-discovery-tcp ws-discovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh
xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-java-gateway zabbix-se
rver zabbix-trapper zabbix-web-service zero-k zerotier
root@server:~# firewall-cmd --add-service=http
success
```

# Изменение обратной DNS-зоны

```

Sep 26
root@server:~ – sudo -i

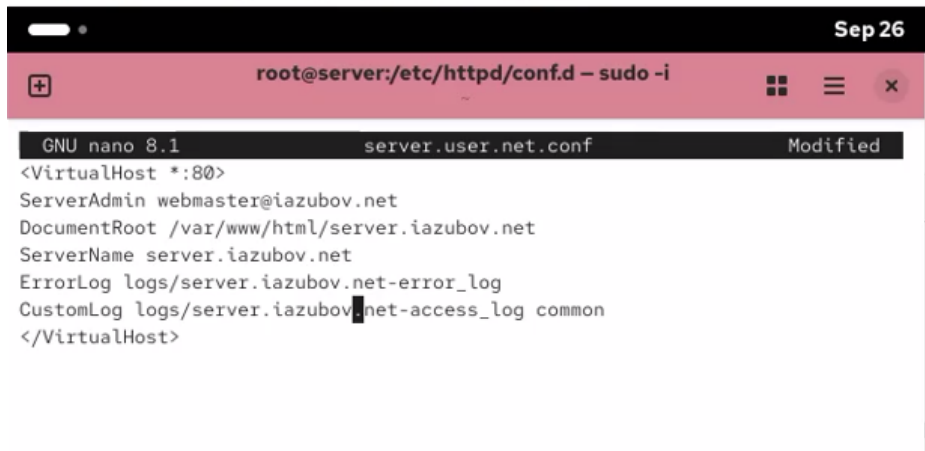
GNU nano 8.1 /var/named/master/rz/192.168.1 Modified
$ORIGIN .
$TTL 86400 ; 1 day
1.168.192.in-addr.arpa IN SOA 1.168.192.in-addr.arpa. server.iazubov.net. (
                                20260926 ; serial
                                86400 ; refresh (1 day)
                                3600 ; retry (1 hour)
                                604800 ; expire (1 week)
                                10800 ; minimum (3 hours)
                                )
                                NS 1.168.192.in-addr.arpa.
                                A 192.168.1.1
                                PTR server.iazubov.net.
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.iazubov.net.
PTR ns.iazubov.net.
PTR dhcp.iazubov.net

$TTL 1200 ; 20 minutes
30 PTR client.iazubov.net.
DHCID ( AAEBD+JNylrBkaDZjmIOjZgXSXbHcGX/jcNYqwqiCn>
IRA= ) ; 1 1 32
1 PTR www.iazubov.net
```

# Удаление журналов DNS и запуск named

```
root@server:~# rm -f /var/named/master/rz/192.168.1.jnl
root@server:~# rm -f /var/named/master/fz/iazubov.net.jnl
root@server:~# systemctl start named
```

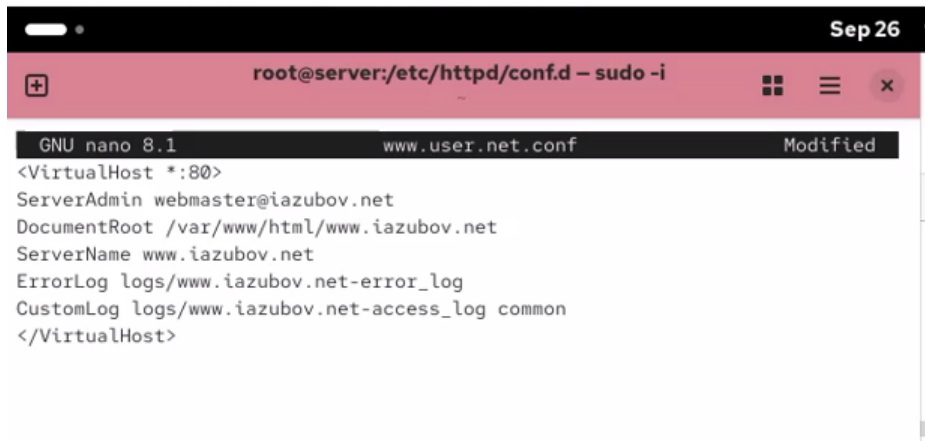
# Создан конфигурационный файл виртуального хоста server.iazubov.net



The screenshot shows a terminal window with a title bar indicating the date "Sep 26". The terminal prompt is "root@server:/etc/httpd/conf.d - sudo -i". The nano text editor is open, editing the file "server.user.net.conf". The editor's status bar at the top shows "GNU nano 8.1", the filename "server.user.net.conf", and the word "Modified". The content of the file is an Apache VirtualHost configuration for \*:80, including the server admin email, document root, server name, and log file locations. The cursor is positioned at the end of the "CustomLog" line.

```
root@server:/etc/httpd/conf.d - sudo -i
GNU nano 8.1 server.user.net.conf Modified
<VirtualHost *:80>
ServerAdmin webmaster@iazubov.net
DocumentRoot /var/www/html/server.iazubov.net
ServerName server.iazubov.net
ErrorLog logs/server.iazubov.net-error_log
CustomLog logs/server.iazubov.net-access_log common
</VirtualHost>
```

# Создан второй виртуальный хост www.iazubov.net



The screenshot shows a terminal window with a pink title bar that reads "root@server:/etc/httpd/conf.d - sudo -i". The window contains the GNU nano 8.1 editor editing the file "www.user.net.conf". The configuration for a new VirtualHost is being entered, with the following content:

```
<VirtualHost *:80>
ServerAdmin webmaster@iazubov.net
DocumentRoot /var/www/html/www.iazubov.net
ServerName www.iazubov.net
ErrorLog logs/www.iazubov.net-error_log
CustomLog logs/www.iazubov.net-access_log common
</VirtualHost>
```

The nano editor interface includes a status bar at the top showing "GNU nano 8.1", the filename "www.user.net.conf", and the word "Modified". The terminal window has a date indicator "Sep 26" in the top right corner.



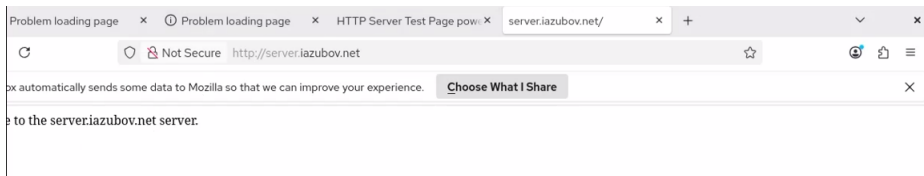
# Создание каталога сайта и index.html

```
root@server:/etc/httpd/conf.d# cd /var/www/html
root@server:/var/www/html# mkdir server.iazubov.net
root@server:/var/www/html# cd /var/www/html/server.iazubov.net
root@server:/var/www/html/server.iazubov.net# touch index.html
root@server:/var/www/html/server.iazubov.net# nano index.html
```

# Права, SELinux и перезапуск Apache

```
root@server:/var/www/html/www.iazubov.net# chown -R apache:apache /var/www
root@server:/var/www/html/www.iazubov.net# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
Relabeled /etc/foomatic/hashes.d/hashes.new from system_u:object_r:etc_runtime_t:s0 to system_u:object_r:etc_t:s0
root@server:/var/www/html/www.iazubov.net# restorecon -vR /var/named
root@server:/var/www/html/www.iazubov.net# restorecon -vR /var/www
root@server:/var/www/html/www.iazubov.net# systemctl restart httpd
root@server:/var/www/html/www.iazubov.net#
```

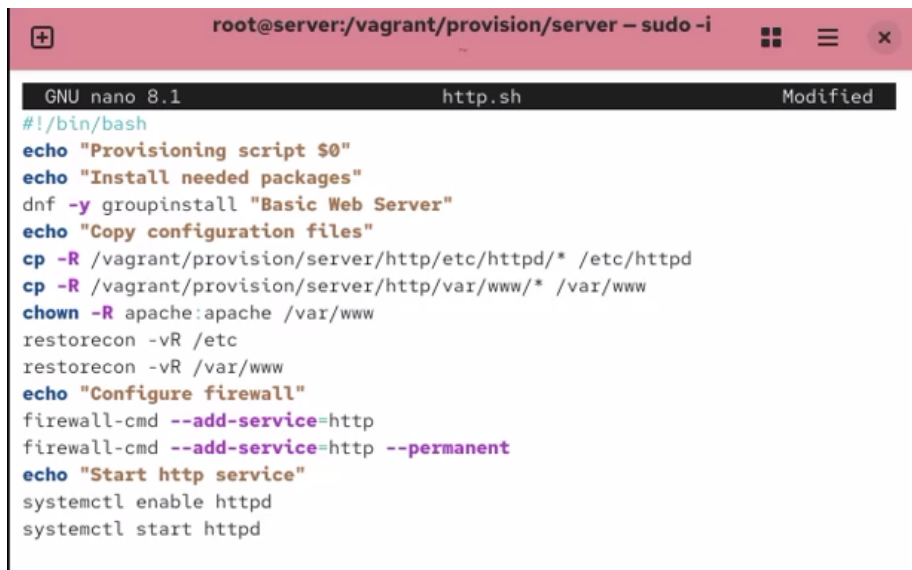
# Проверка виртуального веб-сервера



# Сохранение настроек для Vagrant

```
root@server:/var/www/html/www.iazubov.net# cd /vagrant/provision/server
root@server:/vagrant/provision/server# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
root@server:/vagrant/provision/server# mkdir -p /vagrant/provision/server/http/var/www/html
root@server:/vagrant/provision/server# cp -R /etc/httpd/conf.d/*
/vagrant/provision/server/http/etc/httpd/conf.d/
cp: target '/etc/httpd/conf.d/www.user.net.conf': Not a directory
-bash: /vagrant/provision/server/http/etc/httpd/conf.d/: Is a directory
root@server:/vagrant/provision/server# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d/
root@server:/vagrant/provision/server# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
root@server:/vagrant/provision/server# cd /vagrant/provision/server/dns/
root@server:/vagrant/provision/server/dns# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/fz/iazubov.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/iazubov.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
root@server:/vagrant/provision/server/dns# cd /vagrant/provision/server
```

# Создание http.sh



The screenshot shows a terminal window titled "root@server:/vagrant/provision/server - sudo -i". Inside the terminal, the GNU nano 8.1 editor is open, editing a file named "http.sh". The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
systemctl start httpd
```

# Изменение Vagrantfile



```

type: "shell",
preserve_order: true,
path: "provision/server/dns.sh"
server.vm.provision "server dhcp",
  type: "shell",
  preserve_order: true,
  path: "provision/server/dhcp.sh"
end

server.vm.provision "server dummy",
  type: "shell",
  preserve_order: true,
  path: "provision/server/01-dummy.sh"

server.vm.provision "server http",
  type: "shell",
  preserve_order: true,
  path: "provision/server/http.sh"

```

Стр 74, столб 35    100%    Windows (CRLF)    UTF-8